Covariance Signs for Residue-Class Occupancy on Decimal Unit Carriers

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Abstract

Choose a uniformly random fixed-size subset of the integers below a power of ten that are coprime to ten. We determine the sign of the covariance between the numbers of occupied residue classes modulo 3 and modulo any positive integer coprime to 30. Outside deterministic zero ranges, the covariance is strictly negative except for four families of moduli near half the cutoff. Explicit quadratic expressions determine every exceptional sign; a divisibility argument excludes nontrivial zeros in two of the four families. We also prove a sufficient negative-covariance criterion for arbitrary partitions with three nearly equal rows and controlled column profiles, with a consequence for nonnegative weighted sums over overlapping admissible test sets. The proof combines empty-class identities, a periodic matching count, majorization, and exact polynomial certificates checked by rational interpolation. A minimal standard-library companion verifies all certificate identities. For prime moduli, the classification gives the sign of the covariance of two local sieve factors. It does not assert a distribution law for actual prime sets.

1 Introduction and main results

The integers below 100 that are coprime to 10 form the familiar four-column display

$$\{10j + r : 0 \le j \le 9,\ r \in \{1, 3, 7, 9\}\}.$$

Every prime at least 11 has one of these four terminal digits. The display is therefore a natural carrier for studying local divisibility constraints. Its entries, however, include composite integers and the integer 1. The carrier specifies candidates; it does not specify their primality. This distinction leads to a finite question that can be answered without a prime-distribution hypothesis: when a fixed number of candidate offsets are sampled, how do two counts of occupied residue classes vary together?

Let $d \ge 2$, $N = 10^d$, and

$$\Omega_N = \{h \in \mathbb{Z} : 1 \le h < N,\ \gcd(h, 10) = 1\}, \qquad m = |\Omega_N| = \frac{2N}{5} = 3a + 4, \qquad s = 2a + 4. \tag{1.1}$$

Here

$$a = \frac{4(10^{d-1} - 1)}{3}, \qquad a = 12 \text{ if } d = 2, \quad a \ge 132 \text{ if } d \ge 3.$$

For a uniformly chosen k-element subset H of Ω_N, set

$$\nu_q(H) = |\{h \bmod q : h \in H\}|, \qquad C(d, q, k) = \operatorname{Cov}(\nu_3(H), \nu_q(H)).$$

Unless stated otherwise, $0 \le k \le m$, q is a positive integer, and $\gcd(q, 30) = 1$. The modulus q need not be prime. Binomial coefficients with nonnegative upper index are zero when the lower index is outside its usual range.

Theorem 1 (Decimal sign classification)**.** *If $q = 1$ or $q \geq N/2$, then $C(d, q, k) = 0$ for every k. For $7 \leq q < N/2$, the covariance is zero when $k = 0, 1$ or $k > s$. For $2 \leq k \leq s$, it is strictly negative except possibly when*

$$q = N/2 - g, \qquad g \in \{1, 3, 7\}, \quad \text{or} \quad g = 19 \text{ and } d \geq 3.$$

In these exceptional families put $X = (s - k)(s - k - 1)$. The sign of C, including equality, is the sign of the following expression:

Gap g	*Sign expression*
$1, 3$	$a(5a + 7) - 2X$
7	$(2a + 1)(7a^2 - 7a - 24) - (11a - 2)X$
$19,\ d \geq 3$	$(2a + 1)(a^2 - 49a - 72) - (17a - 14)X$

Each exceptional family starts with negative covariance at $k = 2$ and ends with positive covariance at $k = s - 1, s$. It has one transition from negative to positive and at most one intervening integer zero.

The statement is an exact sign decision for every admitted input. Corollary 11 rules out nontrivial integer zeros for gaps 7 and 19. The corresponding arithmetic question remains open only for gaps 1 and 3; it is separate from deciding the sign at a specified input.

Our second result has no arithmetic carrier in its hypotheses. For a partition, its occupancy is the number of its classes met by the random subset. A column profile records its intersections with the three rows, in the specified row order.

Theorem 2 (A three-row criterion)**.** *Let a finite set have a row partition of sizes $\rho = (a + 2, a + 2, a)$, where $a \geq 40$ is an integer, and a partition into nonempty columns. For each column write $v = b\mathbf{1} + e$, where $\mathbf{1} = (1, 1, 1)$, $b = \min_i v_i$, and $\min_i e_i = 0$. Suppose every column satisfies*

$$b \geq 4, \qquad \max_i e_i \leq 2, \qquad |v| \leq a, \qquad e = (0, 0, 2) \implies b \geq 6. \tag{1.2}$$

For a uniform k-subset, the covariance of the two occupancies is strictly negative for $2 \leq k \leq 2a + 4$, and zero for the other cardinalities $0 \leq k \leq 3a + 4$.

The orientation of $(0, 0, 2)$ is essential: the third row is the smaller one. The criterion is sufficient, with no claim that its numerical thresholds are optimal. It is nonvacuous for infinitely many populations. For example, take $a = 12t$, $t \geq 4$, with $3t - 1$ columns of profile $(4, 4, 4)$ and one column of profile $(6, 6, 4)$. Their sum is $(a + 2, a + 2, a)$ and all hypotheses hold. The argument controls each admissible set separately. Corollary 12 therefore permits overlapping test sets and fixed nonnegative weights, without changing the sampling law.

1.1 Relation to established frameworks

The elementary empty-class formula used below is a finite hypergeometric identity. The occupancy of a partition is also the rank function of the direct sum of rank-one uniform matroids on its classes. Consequently the joint generating function

$$\sum_{H \subseteq \Omega_N} z^{|H|} u^{\nu_3(H)} v^{\nu_q(H)}$$

is an instance of the established rank generating function of a pair of matroids; see Las Vergnas [5, p. 305]. No new general matroid invariant is proposed here. The contribution is the sign classification for this structured pair, and the separate sufficient criterion in Theorem 2.

Negative association is an important framework for sampling without replacement [2]. Its disjoint-coordinate condition does not directly decide our covariance: both occupancies depend on the same sample indicators, and a row event and a column event generally overlap. Indeed Theorem 1 contains positive covariances. Scalar hypergeometric stochastic-order criteria [3] and coupling comparisons for weighted sums [1] concern related sampling questions. The urn comparison needed here changes three population sizes simultaneously and adds three empty-row probabilities; we prove the required comparison explicitly rather than assuming a componentwise stochastic order.

The arithmetic motivation is equally specific. Residue occupancies determine the local factors occurring in singular-series calculations; constrained averages of singular series are studied, for example, by Kuperberg [4]. Section 8 gives the exact two-factor consequence of Theorem 1. An average of an entire singular series, or a statement about actual prime patterns, requires additional information.

1.2 Proof organization and computational content

We first reduce occupancy covariance to empty-class probabilities. Two certified urn inequalities and majorization prove Theorem 2, which controls the small-modulus range. The remaining arithmetic problem splits into a matching window and fourteen finite bands. The matching window gives the exceptional families explicitly. In each band, affine column multiplicities reduce the sign to finitely many polynomial identities.

All load-bearing certificate data are supplied with the article. The verifier uses only integer arithmetic and exact rational arithmetic from Python's standard library. It reconstructs the polynomials and checks identities on complete interpolation grids whose degree bounds are proved below. These checks establish polynomial identities on infinite parameter domains; they are not numerical extrapolations. The certificate specification appears in Section 10. The proof is computer-assisted, not a proof-assistant formalization.

2 Empty-class identities

A complete interval of length 30 contributes four carrier points to each residue class modulo 3. Since $N \equiv 10 \pmod{30}$, the final interval contributes $(2, 2, 0)$. Thus the row sizes are

$$\rho = (a + 2, a + 2, a) \tag{2.1}$$

in the fixed order of residues $0, 1, 2$.

Lemma 3 (Covariance from intersections)**.** *For two partitions of a set of size m, let ρ_r be the row sizes, j_c the column sizes and v_{rc} the intersection sizes. For uniform k-subsets, with $Z = \binom{m}{k}$, their occupancy covariance equals*

$$\sum_{r,c} \left\{ \frac{\binom{m-\rho_r-j_c+v_{rc}}{k}}{Z} - \frac{\binom{m-\rho_r}{k}\binom{m-j_c}{k}}{Z^2} \right\}. \tag{2.2}$$

Proof. Each occupancy is the total number of classes minus the sum of the indicators that a class is empty. The covariance is therefore the sum of the covariances of empty-row and empty-column indicators. Avoiding their union leaves $m - \rho_r - j_c + v_{rc}$ available elements, which proves the formula. □

For a triple $R = (R_0, R_1, R_2)$ of nonnegative population sizes, put

$$F(R; k) = \sum_{r=0}^{2} \frac{\binom{M-R_r}{k}}{\binom{M}{k}}, \qquad M = R_0 + R_1 + R_2, \quad 0 \le k \le M. \tag{2.3}$$

This is the expected number of empty rows. For a column v of size $j = |v|$, conditioning on its being empty gives its contribution to (2.2) as

$$C_v = \frac{\binom{m-j}{k}}{\binom{m}{k}}\big(F(\rho - v; k) - F(\rho; k)\big), \qquad k \le m - j. \tag{2.4}$$

For $k > m - j$ that column is always occupied, so its contribution is zero. Column multiplicities must be retained when summing C_v.

The deterministic boundaries are immediate. When $k = 0, 1$, both occupancies are deterministic. If $k > s = m - a$, every row is occupied. When $q = 1$, the column occupancy is deterministic. Finally, if $q \ge N/2$, two distinct carrier points cannot be congruent modulo q. Both points are odd and q is odd, so their nonzero difference would have magnitude at least $2q \ge N$. Hence $\nu_q(H) = k$.

2.1 A common polynomial kernel

Write $(z)_j = z(z-1)\cdots(z-j+1)$, with $(z)_0 = 1$. For a collection of column profiles choose

$$J \ge \max_v |v|, \qquad D \ge \max_{v,r} \delta_r(v), \qquad \delta_r(v) = |v| - v_r + 2\mathbf{1}_{\{r<2\}},$$

and assume $D \ge 2$. Define

$$\begin{aligned} K_v(a,k) = (m)_J \sum_{r=0}^{2} (s-k)_{\delta_r(v)} \prod_{i=\delta_r(v)}^{D-1} (s-i) \\ - \left(\prod_{i=2}^{D-1}(s-i)\right)\big[s(s-1) + 2(s-k)_2\big](m-k)_{|v|} \prod_{i=|v|}^{J-1}(m-i). \end{aligned} \tag{2.5}$$

Empty products are one. Whenever $2 \le k \le s$, $J \le m$, and $D \le s$, the empty-class identity yields

$$C = \frac{\binom{s}{k}}{\binom{m}{k}(s)_D (m)_J} \sum_v n_v K_v(a,k), \tag{2.6}$$

where n_v is the number of columns with profile v. Indeed $m - \rho_r - |v| + v_r = s - \delta_r(v)$, and

$$\frac{\binom{s-\delta}{k}}{\binom{s}{k}} = \frac{(s-k)_\delta}{(s)_\delta}, \qquad \frac{\binom{m-j}{k}}{\binom{m}{k}} = \frac{(m-k)_j}{(m)_j}.$$

Moreover $F(\rho; k) = \binom{s}{k}\big[s(s-1) + 2(s-k)_2\big]/\big(\binom{m}{k} s(s-1)\big)$. These identities also hold at integer values where the numerator binomial coefficient vanishes.

The raw total degree of K_v in (a, k) is at most $J + D$, and its degree in k is at most $J + 2$. Each first summand has total degree $J + D$ and k-degree $\delta_r(v) \le |v| + 2 \le J + 2$; the second summand has the same total bound and k-degree at most $|v| + 2$.

3 Two finite certificates for urn comparisons

We state the certificate principle explicitly, so that the role of each finite computation is clear.

Lemma 4 (Identity and sign certificates)**.** *Two polynomials of degree at most A in one variable and at most B in another are identical if they agree on a product of $A + 1$ distinct nodes and $B + 1$ distinct nodes. If*

$$Q(t,x) = \sum_{i=0}^{n} B_i(t) \binom{n}{i} x^i (1-x)^{n-i}$$

and each $LB_i(t_0+y)$, for a positive integer L, has nonnegative power coefficients and a strictly positive constant term, then $Q(t,x)>0$ for $t\ge t_0$, $0\le x\le 1$. The analogous statement with $-LB_i$ gives strict negativity.

Proof. Apply the univariate root bound successively in the two variables. For the sign statement, every coefficient polynomial is strictly positive for $y\ge 0$, and the Bernstein basis is nonnegative and has sum one on $[0,1]$. □

Lemma 5 (Balanced urn increment)**.** *Let $R=(h,h+p,h+r)$, where $h\ge 16$ and $0\le p\le r\le 4$ are integers. In the common domain $0\le k\le 3h+p+r$,*

$$F(R+\mathbf{1};k)\ge F(R;k).$$

The inequality is strict for $2\le k\le 2h+p+r+2$, and is an equality for the other cardinalities in that domain.

Proof. Put $M=3h+p+r$, $n=2h+p+r$, $D=r+2$. The list $\mathcal{D}=(0,p,r)$ retains repetitions. Set

$$\begin{aligned}A&=\sum_{d\in\mathcal{D}}(n+2-k)_d\prod_{i=d}^{D-1}(n+2-i),\\ B&=\sum_{d\in\mathcal{D}}(n+2-k)_{d+2}\prod_{i=d+2}^{D-1}(n+2-i),\\ P&=(M+3-k)_3A-(M+3)_3B.\end{aligned}$$

Conversion of the binomial coefficients in (2.3) gives

$$F(R+\mathbf{1};k)-F(R;k)=\frac{\binom{n+2}{k}}{\binom{M+3}{k}}\frac{P}{(n+2)_D(M+3-k)_3}\tag{3.1}$$

for $2\le k\le n+2$. Every factor outside P is positive in this range: $n+2\ge D$ and $M+3-k\ge h+1\ge 17$.

For each of the fifteen pairs (p,r), the certificate `urn_monotonicity.json` establishes

$$P=k(k-1)Q,\qquad Q\mid_{k=2+(2h+p+r)x}=\sum_{i=0}^{r+1}B_i(h)\binom{r+1}{i}x^i(1-x)^{r+1-i}.$$

It records the coefficients of $LB_i(16+y)$ for a positive integer scale L. They are nonnegative and their constants are positive. The parameter degree bound of the stored coefficient polynomials is $r+3$; there are 490 stored integers in total.

For completeness, the raw degree bounds follow from the displayed formula for P: its total degree in (h,k) is at most $r+5$, and its k-degree is at most $r+3$. After $k=2+(2h+p+r)x$, its degree in h remains at most $r+5$, and its degree in x at most $r+3$. The encoded right side has these same bounds. The checker reconstructs both sides and compares them at $r+6$ values of h, starting at 16, and $r+4$ values $x=j/(r+3)$. All 890 equalities hold. Lemma 4 therefore proves the identities globally and $Q>0$ on the required domain. Equation (3.1) proves strictness.

For $k>n+2$, both empty-row expectations vanish. For $k=0,1$, they are respectively $3,2$, proving the remaining equalities. □

Lemma 6 (Six column anchors)**.** *Let $\rho=(a+2,a+2,a)$, $a\ge 40$. For $2\le k\le 2a+4$,*

$$F(\rho-v;k)<F(\rho;k)$$

for each of the ordered vectors

$$v\in\{(4,4,4),(4,4,5),(4,5,5),(4,5,6),(4,6,6),(6,6,8)\}.\tag{3.2}$$

Proof. For each anchor use (2.5) with $J = |v|$ and $D = \max_r \delta_r(v)$. Then

$$\frac{\binom{m-J}{k}}{\binom{m}{k}}\big(F(\rho - v; k) - F(\rho; k)\big) = \frac{\binom{s}{k}}{\binom{m}{k}}\frac{K_v(a,k)}{(s)_D(m)_J}. \tag{3.3}$$

The certificate `column_anchors.json` establishes $K_v = k(k-1)Q_v$. After $k = 2 + (2a+2)x$, it gives the degree-J Bernstein expansion of Q_v, whose coefficient polynomials have degree at most $J + D - 2$ in a. For each coefficient $B_i(a)$, the power coefficients of $-LB_i(40 + y)$ are nonnegative with positive constant term. Thus the *Bernstein coefficients themselves* are strictly negative for $a \geq 40$.

There are 2630 stored integers. The raw bounds $J + D$ in total degree and $J + 2$ in k follow from Section 2.1. They give bounds $J + D, J + 2$ after the substitution. The encoded right side has the same bounds. The checker verifies the raw identity on $(J + D + 1)(J + 3)$ rational grid points for each anchor, totalling 3166 exact equalities. Lemma 4 therefore proves the global identity and strict negativity. All denominators in (3.3) are positive. Also $J \leq 20 \leq a$, so $m - J \geq s$ and its left prefactor is positive throughout the asserted range. □

4 Proof of the three-row criterion

At fixed total size M, the function $F(R; k)$ is Schur-convex on integer triples: making the row sizes more unequal increases it weakly. This follows directly from discrete convexity, since

$$\binom{M-z-2}{k} - 2\binom{M-z-1}{k} + \binom{M-z}{k} = \binom{M-z-2}{k-2} \geq 0 \tag{4.1}$$

whenever the indicated row-size comparison is defined. Equivalently, a unit transfer from a smaller entry to a larger one cannot decrease the sum in (2.3).

Proof of Theorem 2. There are nineteen possible normalized profiles $e \in \{0,1,2\}^3$ with $\min e = 0$. For each choose a comparison profile e^* and an anchor base b^* from Table 1.

Profile class	e^*	b^*	$b^*\mathbf{1} + e^*$
$\lvert e\rvert = 0$	$(0,0,0)$	4	$(4,4,4)$
$\lvert e\rvert = 1$	$(0,0,1)$	4	$(4,4,5)$
$\lvert e\rvert = 2,\ e \neq (0,0,2)$	$(0,1,1)$	4	$(4,5,5)$
$\lvert e\rvert = 3$	$(0,1,2)$	4	$(4,5,6)$
$\lvert e\rvert = 4$	$(0,2,2)$	4	$(4,6,6)$
$e = (0,0,2)$	$(0,0,2)$	6	$(6,6,8)$

Table 1: Comparison profiles for the three-row criterion.

The vector $(2 - e_0^*, 2 - e_1^*, -e_2^*)$ majorizes $(2 - e_0, 2 - e_1, -e_2)$: their sums agree, the former minimum is no larger, and its maximum is no smaller. These are precisely the two majorization inequalities for triples of equal sum. The table can also be verified over the nineteen profiles by the companion. Adding the common value $a - b$ gives

$$F(\rho - b\mathbf{1} - e; k) \leq F(\rho - b\mathbf{1} - e^*; k).$$

Since $3b \leq |v| \leq a$, the minimum of either residual population is at least

$$a - b - 2 \geq 2a/3 - 2 > 16,$$

and its spread is at most four. By symmetry of F, the entries may be sorted before applying Lemma 5. Decreasing b one unit at a time until $b = b^*$ adds $\mathbf{1}$ at each step. The hypotheses give $b \geq b^*$, so

$$F(\rho - v; k) \leq F(\rho - b\mathbf{1} - e^*; k) \leq F(\rho - b^*\mathbf{1} - e^*; k) < F(\rho; k).$$

The final strict inequality is Lemma 6. The fixed k lies in the domain of every intermediate urn, because the initial remaining population has size $m - |v| \geq s$. Where necessary the increment lemma is used weakly, including ranges where both empty-row expectations vanish.

For $2 \leq k \leq s$, the prefactor in (2.4) is positive for every column, again because $m - |v| \geq s$. Every contribution is strictly negative, so their sum is strictly negative. The zero ranges follow from Section 2. □

Remark 7 (Why the orientation restriction is retained). For $a = 40$, $k = 2$, the column $v = (4, 4, 6)$ has a *positive* contribution $C_v = 71/4846323$. This can be checked by counting pairs: put $I = \mathbf{1}_{H \cap B \neq \varnothing}$ and $D = \binom{124}{2} = 7626$. Across these pairs, $\sum U = 12750$, $\sum I = 1631$ and $\sum UI = 2727$; hence

$$C_v = \frac{2727}{7626} - \frac{12750 \cdot 1631}{7626^2} = \frac{71}{4846323} > 0.$$

Thus the condition on the normalized profile $(0, 0, 2)$ cannot be removed from this columnwise proof. This is a counterexample to the individual-column inequality, not a counterexample for an entire partition satisfying weakened hypotheses. Similarly a completely unrestricted balanced urn increment is false:

$$F((2, 6, 6); 2) - F((1, 5, 5); 2) = -\frac{23}{1001}.$$

5 Small moduli from periodic column profiles

The criterion has an application slightly wider than the decimal carrier.

Proposition 8. *Let $N \equiv 10 \pmod{30}$, $q \geq 7$, $\gcd(q, 30) = 1$, and $N > 32q$. Use the carrier Ω_N, let $m = 2N/5 = 3a + 4$, and sample uniformly among its k-subsets. Then*

$$\operatorname{Cov}(\nu_3, \nu_q) < 0 \quad (2 \leq k \leq 2a + 4),$$

and the covariance is zero for the other admissible cardinalities.

Proof. Each class modulo the odd modulus q has a unique odd representative $h \in [1, 2q)$. Its odd terms below N are $h, h + 2q, \ldots$; remove the terms divisible by 5. Since $N > 32q$, each such progression has at least sixteen terms.

Any fifteen consecutive terms traverse all odd residues modulo 30. Deleting the multiples of 5 leaves four terms in each row modulo 3. Thus adding a full period adds $(4, 4, 4)$, and $b = \min v \geq 4$. Before deletion, the three row counts differ by at most one. Deleted indices have step five; their row counts also differ by at most one. Hence the surviving row counts have spread at most two.

It remains to check the oriented exception in (1.2). Remove all full periods, leaving a residual progression of length $0 \leq L \leq 14$. For its normalized profile to be $(0, 0, 2)$, row 2 must have one extra term before deletion and one fewer deleted term than each other row. Consequently $L \in \{1, 4, 7, 10, 13\}$, the progression starts in row 2, and there are exactly two deleted terms, neither in row 2. Indeed the pre-deletion counts can differ by only one, the deletion counts by only one, and at most three deletions are possible in fourteen terms.

Write the deleted indices as $c, c+5, \ldots$, $0 \le c \le 4$. Lengths $1, 4$ allow at most one deletion. For length 7, two deletions require $c = 0$ or 1, forcing a deleted row-2 index at 0 or 6. For length 13, exactly two require $c = 3$ or 4, forcing a deleted row-2 index at 3 or 9. At length 10, only $c = 2$ avoids row 2, leaving $(2, 2, 4)$. Restoring full periods gives

$$v = (2 + 4z, 2 + 4z, 4 + 4z).$$

The original length is at least sixteen, so $z \ge 1$ and $b \ge 6$.

Finally,

$$|v| \le \left\lceil \frac{N}{2q} \right\rceil \le \frac{15a + 48}{28} \le a \qquad (a \ge 4),$$

where the middle inequality uses $q \ge 7$, $N = (15a + 20)/2$, and $\lceil x \rceil \le x + 1$. For $a \ge 40$, Theorem 2 applies.

If $a < 40$, the constraints $N \equiv 10 \pmod{30}$, $224 < N < 310$, and $N > 32q$ leave only $(N, q) = (250, 7), (280, 7)$. Their 101 and 113 possible cardinalities are checked exactly by (2.2); the fractions are in `small_cutoffs.json`. They are strictly negative in the asserted range and zero outside it. □

For $N = 10^d$, these finite extended exceptions are unnecessary: at $d = 2$ there is no admissible $7 \le q < N/32$, while at $d \ge 3$ one has $a \ge 132$.

6 The matching window and exceptional families

Suppose $N/4 \le q < N/2$, and put $t = N - 2q$. Every nonsingleton column is a pair with difference $2q$, since three odd terms would span at least $4q \ge N$. The pairs are exactly

$$\{h, h + 2q\}, \qquad 1 \le h < t, \qquad \gcd(h(t - h), 10) = 1. \tag{6.1}$$

Let M be their number and let γ count their endpoints in row 2. No pair meets a row twice, because $3 \nmid q$. Reflection $h \mapsto N - h$ exchanges rows $0, 1$ and fixes row 2, so the numbers of pairs of each type are

$$e_{01} = M - \gamma, \qquad e_{02} = e_{12} = \gamma/2.$$

The pair centered at $N/2$, with endpoints $N/2 - q, N/2 + q$, has both endpoints coprime to 10; hence $M \ge 1$.

Put

$$b = 2a + 2, \quad T = m(m - 1), \quad B = b(b - 1), \quad S = s(s - 1), \tag{6.2}$$

and

$$L = T\gamma - 2MB, \quad V = M(T - S) - T\gamma, \quad P(k) = L(s - k)(s - k - 1) + BV. \tag{6.3}$$

Lemma 9. *For $2 \le k \le s$,*

$$C(d, q, k) = \frac{k(k - 1)\binom{s}{k}}{\binom{m}{k} STB} P(k). \tag{6.4}$$

Moreover

$$P(2) = -MB(3a^2 + 5a + 4) < 0, \qquad P(s - 1) = P(s) = BV.$$

Proof. Let E_r indicate an empty row and I_e the event that both endpoints of pair e are selected. Then $\nu_3 = 3 - \sum_r E_r$ and $\nu_q = k - \sum_e I_e$. Counting pairs wholly outside each row yields

$$C = \frac{\gamma\binom{b-2}{k-2} + (M - \gamma)\binom{s-2}{k-2}}{\binom{m}{k}} - \frac{Mk(k - 1)}{T\binom{m}{k}} \left(2\binom{b}{k} + \binom{s}{k} \right).$$

Use

$$\binom{n-2}{k-2} = \binom{n}{k}\frac{k(k-1)}{n(n-1)}, \qquad \binom{b}{k} = \binom{s}{k}\frac{(s-k)(s-k-1)}{S}$$

to obtain (6.4). Substitution at $k = 2, s-1, s$ gives the endpoint formulas. □

The restriction $\gcd(q, 30) = 1$ makes $t = r + 60j$, $j \geq 0$, with base counts as in Table 2.

r	M_r	γ_r	$D_r = 2M_r - 3\gamma_r$
2	1	0	2
6	1	0	2
14	5	2	4
18	5	4	-2
26	7	4	2
38	11	6	4
42	13	8	2
54	17	12	-2

Table 2: The periodic pair count. For $t = r + 60j$, $M = M_r + 18j$ and $\gamma = \gamma_r + 12j$.

The base counts follow directly from (6.1). Over each interval of sixty possible starts there are eighteen admissible starts, six in each residue class modulo 3: admissibility allows three residues modulo 10. Since the two endpoints occupy distinct rows, these starts give twelve additional endpoints in row 2. This proves the period formula, without extrapolation from samples.

Since $X = (s-k)(s-k-1)$ decreases to zero, P is affine in X. If $V < 0$, both endpoint values of P are negative, so all its values in the range are negative. To have $V \geq 0$, one must have

$$\frac{\gamma}{M} \leq \frac{T-S}{T} = \frac{5a^2+7a}{9a^2+21a+12} < \frac{5}{9}. \tag{6.5}$$

Consequently $D_r = 2M - 3\gamma > M/3$, or $M < 3D_r$. Table 2 leaves only $t = 2, 6, 14, 38$. For $t = 2, 6$, $V > 0$; for $t = 14$,

$$V = 7a^2 - 7a - 24 > 0 \qquad (a \geq 12);$$

for $t = 38$,

$$V = a^2 - 49a - 72,$$

which is negative at $a = 12$ and positive at every $a \geq 132$. Thus $V = 0$ never occurs at a decimal depth.

Substituting these four pair counts into (6.3) and dividing by positive factors gives exactly the three sign expressions in Theorem 1. For $t = 2, 6$ the factor is B; for $t = 14, 38$ it is $2(a+1)$. Their coefficients of X are negative. The sequence X is strictly decreasing up to $k = s-1$, and equals zero at both $s-1, s$, where the sign is positive. This proves the single transition and the bound of one integer zero.

7 Intermediate moduli by affine bands

We now treat $N/32 \leq q < N/4$. For $d \geq 3$, the proof reduces to fourteen finite certificate families.

7.1 Column multiplicities

Put

$$\ell = \left\lceil \frac{N}{2q} \right\rceil, \qquad \frac{N}{2\ell} \le q < \frac{N}{2(\ell-1)}, \qquad 3 \le \ell \le 16.$$

For the odd representative $h \in [1, 2q)$ of a class, the odd progression has length ℓ when $h < t = N - 2(\ell-1)q$, and length $\ell - 1$ otherwise. Fix $u = q \bmod 30$ among the eight units modulo 30, and put

$$t_0 = (10 - 2(\ell-1)u) \bmod 30, \qquad b_0 = 2u \bmod 30.$$

For each odd $r \in \{1, 3, \dots, 29\}$, the numbers of representatives in the two length ranges are

$$w_{\ell,r} = \frac{N - 2(\ell-1)q - t_0}{30} + \mathbf{1}_{\{r<t_0\}}, \tag{7.1}$$

$$w_{\ell-1,r} = \frac{-N + 2\ell q + t_0 - b_0}{30} + \mathbf{1}_{\{r<b_0\}} - \mathbf{1}_{\{r<t_0\}}. \tag{7.2}$$

The column profile from a length L and a start r is

$$v_i^{(L,u,r)} = |\{0 \le j < L : r + 2ju \equiv i \pmod 3,\ 5 \nmid r + 2ju\}|. \tag{7.3}$$

There are thirty length/start possibilities. Aggregating their weights gives

$$n_v = \frac{A_{\ell,u,v} N + B_{\ell,u,v} q + C_{\ell,u,v}}{30}. \tag{7.4}$$

The checker reconstructs these integer triples from (7.1)–(7.3). No precomputed column multiplicity is trusted without comparison.

7.2 Sign certificates

For a fixed band take $J = \max_v |v|$ and $D = \max_{v,r} \delta_r(v)$, over all its residue cases. The numerator of (2.6), multiplied by 30, is

$$P_{\ell,u}(a, q, k) = \sum_v (A_{\ell,u,v} N + B_{\ell,u,v} q + C_{\ell,u,v}) K_v(a, k), \qquad N = (15a + 20)/2. \tag{7.5}$$

It is affine in q. At each formal endpoint $q = N/(2\ell)$ and $q = N/(2(\ell-1))$, the certificate establishes

$$P_{\ell,u} = k(k-1) R_{\ell,u}, \qquad R_{\ell,u}\,\big|_{k=2+(2a+2)x} = \sum_{i=0}^{J} B_i(a) \binom{J}{i} x^i (1-x)^{J-i}. \tag{7.6}$$

For a positive integer scale L, it records every coefficient of $-L B_i(132 + y)$, of degree at most $J + D - 1$. All 35 664 stored integers are strictly positive. It follows that $B_i(a) < 0$ for every $a \ge 132$. By Lemma 4, (7.6) is strictly negative at each endpoint for $2 \le k \le s$. Affinity in q proves the same strict sign between them.

These endpoints are evaluations of formal polynomials. They need not satisfy the residue condition $q \equiv u \pmod{30}$, and their column weights need not describe an actual partition. Affinity requires only the two polynomial endpoint values.

The raw total degree of an endpoint polynomial in (a, k) is at most $J + D + 1$, and its k-degree is at most $J + 2$, by Section 2.1 and the affine multiplicity factor. After the substitution for k, the degree bounds in (a, x) are therefore $J + D + 1, J + 2$. The stored right side of (7.6), including $k(k-1)$, has these same separate bounds. Equality at $J + D + 2$ consecutive a-values starting at 132 and $J + 3$ nodes $x = j/(J + 2)$ proves the full identity. In particular, the factor $k(k-1)$ and the Bernstein representation are verified, not merely inferred from vanishing at a few sample sizes.

Table 3 lists all dimensions. Since $J \le 13$, $D \le 11$, and $a \ge 132$, the denominator in (2.6) is positive. Thus the verified negative numerator proves negative covariance.

ℓ	J	D	Stored integers	Grid equalities
3	3	4	448	864
4	4	5	720	1232
5	4	6	800	1344
6	5	6	1056	1664
7	6	7	1456	2160
8	7	7	1792	2560
9	8	8	2304	3168
10	8	8	2304	3168
11	9	9	2880	3840
12	10	9	3344	4368
13	11	10	4032	5152
14	12	10	4576	5760
15	12	10	4576	5760
16	13	11	5376	6656
Total			35664	47696

Table 3: Complete band certificates. Grid counts include all eight unit residues and both formal endpoints.

7.3 The finite depth and the final join

At $d = 2$, the moduli $7 \le q < N/4$ coprime to 30 are $7, 11, 13, 17, 19, 23$. For these six moduli, the 246 fractions from (2.2), covering all 41 cardinalities, are in `depth2.json`. The checker recomputes each fraction and compares it with the stored value. They are strictly negative for $2 \le k \le 28$ and zero otherwise.

Completion of Theorem 1. Section 2 proves every deterministic zero. Section 6 proves the classification on $N/4 \le q < N/2$. For $d \ge 3$, Section 7 proves negativity on $N/32 \le q < N/4$, while Proposition 8 proves it for $7 \le q < N/32$. The first interval includes its lower endpoint and the second uses a strict upper endpoint, so the join has no gap. The exhaustive depth-two calculation completes the remaining cases. □

8 The arithmetic meaning: two local sieve factors

For fixed $H \subseteq \Omega_N$, consider the translated block forms

$$Nb + h, \qquad h \in H, \qquad b \in \mathbb{Z}.$$

If p is prime and $p \nmid 10$, multiplication by N is invertible modulo p. The set of block indices $b \bmod p$ for which at least one form is divisible by p therefore has exactly $\nu_p(H)$ elements. Consequently

$$L_p(H) = 1 - \frac{\nu_p(H)}{p} \tag{8.1}$$

is the exact fraction of block indices avoiding this local obstruction. This assertion is a finite residue count, with no distributional assumption about primes.

Corollary 10. *For the uniform k-subset model and a prime $q \ge 7$,*

$$\mathrm{Cov}_H(L_3(H), L_q(H)) = \frac{C(d, q, k)}{3q}. \tag{8.2}$$

Equivalently,

$$\mathbb{E}_H[L_3 L_q] = \mathbb{E}_H[L_3]\mathbb{E}_H[L_q] + \frac{C(d,q,k)}{3q}.$$

Thus Theorem 1 gives the sign of the exact correction to the product of these two mean local factors.

Proof. Both local factors are affine functions of their occupancies. Their covariance is the product of the two slopes times $\mathrm{Cov}(\nu_3, \nu_q)$. □

For fixed k, division of each local factor by $(1-1/p)^k$ is a positive deterministic normalization, so it preserves the sign in (8.2). Such normalized factors are the ones appearing in the usual singular-series framework [4]. For the block forms here, the primes $2, 5$ divide no form and give constant local root counts zero. One must not replace this block model by unrestricted translates $b + h$: their local factors at $2, 5$ are different.

For example, at $N = 100$, every prime $p > 50$ acts injectively on the carrier, so $\nu_p(H) = k$. Shape dependence of the block local factors is confined to the thirteen primes

$$3, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.$$

This finiteness does not turn the two-factor covariance theorem into a formula for their entire product.

There is also a concrete obstruction to substituting actual primes for uniform subsets. A uniform subset can occupy all three residue classes modulo 3, giving $L_3 = 0$. A set of offsets whose forms are all primes greater than 3 in one fixed block must avoid the forbidden class modulo 3, so it has $\nu_3 \le 2$. The sampling law in this article is not the law of those prime sets. No prime prediction, prime-pattern asymptotic, or bound on prime gaps is deduced.

9 Consequences, examples and further questions

9.1 Strict transitions for two exceptional gaps

Corollary 11 (No nontrivial zeros for gaps 7 and 19)**.** *For $2 \le k \le s$, the covariance in Theorem 1 does not vanish when $q = N/2 - 7$, $d \ge 2$, or when $q = N/2 - 19$, $d \ge 3$. In each family its transition from negative to positive is strict at integer sample sizes. The deterministic zero ranges are unchanged.*

Proof. Write $f_7(a) = (2a+1)(7a^2 - 7a - 24)$ and $D_7 = 11a - 2$. Since $X = (s-k)(s-k-1)$ is integral, a zero would require $D_7 \mid f_7(a)$. The polynomial identity

$$11^3 f_7(a) = (11a - 2)(1694a^2 - 539a - 6753) - 45450$$

then forces $D_7 \mid 45450$. For $d \ge 5$, one has $D_7 \ge 146650 > 45450$. At $d = 2, 3, 4$, its values are $130, 1450, 14650$, and the remainders of 45450 are respectively $80, 500, 1500$. Thus no admitted depth satisfies the necessary divisibility.

For gap 19, put $f_{19}(a) = (2a+1)(a^2 - 49a - 72)$ and $D_{19} = 17a - 14$. Now

$$17^3 f_{19}(a) = (17a - 14)(578a^2 - 27557a - 78471) - 1452330,$$

so a zero requires $D_{19} \mid 1452330$. For $d \ge 6$, $D_{19} \ge 2266630 > 1452330$. At $d = 3, 4, 5$, the divisors are $2230, 22630, 226630$, with remainders $600, 4010, 92550$. None divides the constant. The endpoint signs and single transition follow from Theorem 1. □

For gaps 1 and 3, put $j = s - k$. The common zero equation $2j(j-1) = a(5a+7)$ has an admissible integer solution exactly when

$$D_d = 1 + 10a^2 + 14a \tag{9.1}$$

is a square. If so, its square root is odd and $j = (1+\sqrt{D_d})/2$; the endpoint signs in Theorem 1 place this root in the required interval. For $n = d - 1 \geq 1$, the same condition is equivalent to the existence of an integer y satisfying

$$y^2 = 160\,10^{2n} - 152\,10^n + 1.$$

Indeed, the right-hand side equals $9D_d$, so every such y is divisible by 3. We leave this exponential Diophantine question open.

9.2 Overlapping admissible test sets

Corollary 12 (Nonnegative weighted occupancy)**.** *Let $E = R_0 \sqcup R_1 \sqcup R_2$, where $(|R_0|, |R_1|, |R_2|) = (a+2, a+2, a) = \rho$, for an integer $a \geq 40$. For a uniform k-subset $H \subseteq E$, let $U(H)$ count the rows meeting H. Let $B_1, \ldots, B_t \subseteq E$ be a finite nonempty family whose oriented intersection vectors*

$$v_i = (|B_i \cap R_0|, |B_i \cap R_1|, |B_i \cap R_2|) = b_i\mathbf{1} + e_i, \qquad b_i = \min_r (v_i)_r,$$

each satisfy (1.2)*. For fixed real weights $w_i \geq 0$ with $\sum_i w_i > 0$, one has*

$$\operatorname{Cov}\left(U(H), \sum_{i=1}^{t} w_i \mathbf{1}_{\{H \cap B_i \neq \varnothing\}}\right) < 0 \qquad (2 \leq k \leq 2a+4).$$

The covariance is zero at every other allowed cardinality $0 \leq k \leq 3a+4$. The sets B_i may overlap and need not cover E.

Proof. Fix an admissible B with intersection vector v, and let $m = 3a+4$, $j = |B| = |v|$. For $2 \leq k \leq s = 2a+4$, the avoidance identity (2.4) gives

$$\operatorname{Cov}(U(H), \mathbf{1}_{\{H \cap B \neq \varnothing\}}) = \frac{\binom{m-j}{k}}{\binom{m}{k}} \big(F(\rho - v; k) - F(\rho; k)\big).$$

Its derivation uses only one set B, not a column partition. The argument in the proof of Theorem 2, before summing over columns, establishes $F(\rho - v; k) < F(\rho; k)$ for each vector satisfying (1.2). In particular, the orientation of $(0, 0, 2)$ and its extra bound on b are retained. The prefactor is positive since $m - j \geq 2a + 4 \geq k$. Every positively weighted term therefore has negative covariance; linearity gives the assertion. For $k = 0, 1$, or $k > 2a+4$, the row count U is deterministic. □

Taking the B_i to be a column partition and all weights equal to one recovers Theorem 2. Alternatively, weights may be assigned only to admissible columns of a larger partition. No signed-weight, stochastic negative-association, or nonuniform sampling statement is asserted.

9.3 Examples and remaining questions

The sign changes are already visible in the smallest decimal carrier:

$$C(2, 47, 8) = \frac{1771}{20974005} > 0.$$

At a common cardinality $k = 15$, one has

$$C(2, 41, 15) = -\frac{51037}{134914728} < 0, \qquad C(2, 43, 15) = \frac{161}{10939032} > 0.$$

The row sizes are the same, and both column partitions have five pairs and thirty singletons, hence the same marginal law for ν_q. Their intersections with the three rows differ. Thus marginal occupancy laws alone cannot determine the covariance.

The matroid interpretation also has a limit. For $N = 100, q = 7$, the two rank functions are not a matroid perspective in either direction. Adding 17 to $\{3\}$ increases the modulo-3 rank by one and the modulo-7 rank by zero. Adding 7 to $\{1\}$ does the reverse. These two examples violate the respective rank-increment inequalities for a perspective [5, p. 304]. The sign theorem is therefore not presented as an application of a perspective-specific positivity statement.

Several questions remain distinct from the proved results. The thresholds $h \geq 16$ and $a \geq 40$ in the urn inequalities may admit improvement, but sharpness is not needed here. An elementary proof eliminating their coefficient certificates would shorten the computational dependency. After Corollary 11, the integer-zero question is restricted to gaps 1 and 3, equivalently the square condition (9.1). Other bases and other pairs of moduli require separate analysis. Finally, a useful transfer to prime patterns would need a controlled full-product average or a justified nonuniform law on offset sets. Pairwise covariance supplies only one part of that problem.

10 Certificate specification and reproduction

The companion folder is self-contained and uses Python 3.10 or later. It contains eighteen JSON certificate files: fourteen bands, two urn files, and two finite-boundary files. A manifest pins all twenty-six payload files by byte count and SHA-256; the manifest itself is the twenty-seventh file. The original certificate bytes are retained, with their source-member identities in `PROVENANCE.json`. The checker requires no original ZIP, symbolic generator, historical prime dataset, or internal project document.

10.1 Encodings and independent reconstruction

For a band, the certificate declares ℓ, J, D, a positive scale, the shift 132, affine type tables, endpoint denominators and every Bernstein index. The checker derives J, D, the type alphabet and the affine multiplicities independently from (7.3). It requires exactly all eight residues and both endpoints. The integer array for index i lists the coefficients of $-LB_i(132 + y)$ in ascending powers of y.

For the urn and anchor files, the arrays are the coefficients of $\sigma LB_i(t_0 + y)$, again in ascending powers. Here $(\sigma, t_0) = (1, 16)$ for the urn increment and $(-1, 40)$ for the anchors. Thus the reconstructed Bernstein coefficient is σ/L times the stored power polynomial. Array dimensions, positive scale, domain shift, sign, and positive constant terms are checked explicitly. The raw polynomials come from the displayed formulas in the proofs, not from the symbolic generator that produced the arrays.

The complete identity workload is

$$47\,696 + 890 + 3166 = 51\,752$$

exact rational equalities, using 38 784 stored integer coefficients. In addition, the finite boundaries require $246 + 214 = 460$ exact fractions. Direct probability comparisons and adverse examples provide separate checks of the algebra-to-probability conversion. They supplement, rather than replace, the global polynomial identities.

10.2 How to run and interpret the checks

From the companion directory, run

```
python -B verify.py
python -B -O verify.py
python -B -m unittest -v test_companion
python -B -O -m unittest -v test_companion
```

The verifier writes nothing and prints a deterministic JSON receipt. If output is redirected, the destination should be outside the companion directory, whose inventory is checked exactly. The normal and optimized receipts must agree byte for byte. The suite contains 17 unit tests: 16 negative controls and one positive inventory check. The test suite checks rejection of omitted cases, corrupted coefficients, altered geometry, weakened degree declarations, wrong domains and missing files; it does not rely on Python assertions being enabled. A clean copy of the folder suffices to repeat every check.

The certificate map identifies each proof dependency and grid size. Hashes establish consistency with that particular package, not a trusted digital signature. A successful checker run does not prove that an arbitrary revised manuscript correctly invokes the identities: the written domain, positivity and reduction arguments remain part of the mathematical proof.

Preparation note

AI assistance contributed to mathematical exploration, proof development, code preparation, verification and drafting. The author supplied research packages produced with an external AI agent; its exact model and version are not documented in the available record. Integration, manuscript preparation and computational checks used the Codex assistant. A separate AI reading examined an earlier version. This is not human peer review, and complete human verification is not claimed here. The companion records certificate provenance and checker adaptations. Exact computational replay supports the stated identities; it does not establish novelty or replace review of the written arguments.

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